

Burning worlds of cartography: a critical approach to climate cosmograms of the Anthropocene

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Climate science today makes use of a variety of red globes to explore and communicate findings. These transform the iconography which informs this image: the idealised, even mythical vision of the blue, vulnerable and perfect marble is impaired by the application of the colours yellow and red. Since only predictions that employ a lot of red seem to exist, spectators are confronted with the message that the future Earth that might turn out as envisaged here is undesirable. Here intuitively powerful narrations of the end of the world may connect. By employing methods of art history and visual analysis, and building on examples from current Intergovernmental Panel on Climate Change reports and future scenario maps, this article explores how burning world images bear – intentionally or not – elements of horror and shock. My question explored here is as follows: should ‘burning world’ images be understood as a new and powerful cosmology?

Key words cartography; visualisation; climate change; whole Earth images

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Introduction

Because of their expressive power, scientific images of climate change are employed to bridge cultural and disciplinary boundaries. Since maps and graphs depicting scenarios of climate change futures do not only follow the normative ideal of signifying rational numbers, they also affect emotion in that they trigger very different and conflicting modes of perception causing tensions between scientific and public meanings. Though the graphs are embedded in complex scientific processes, they still cause an emotional reaction comparable to shock images (Barthes 1972).

In September 2013, the Intergovernmental Panel on Climate Change (IPCC) presented its fifth report on climate change during a press conference in Stockholm. On this occasion, the audience was also presented with some of the latest scientific images produced for the new *Summary for policy makers*. In the manner of previous reports, coloured global maps, barren graphs and curves served as most simplified condensates for visually communicating the findings of the most complex research field of climate change.

For the fifth report, it seemed the graphs had been created in a more straightforward, simplified manner to put across the urgency of the messages delivered to decisionmakers even more clearly (Hollin and Pearce 2015). Thomas Stocker, Professor of Climate and Environmental Physics and head of the IPCC Working Group I focusing on the physical science basis of climate change, directly addressed the imaginative power of scientific imagery. He declared, while pointing to a global map marking mean temperatures in bright red (Figure 1a): ‘This is the face of the surface of our planet if you look at the atmosphere. It is red. The world has been warming. The trend that you see is clearly given as colours of red’.¹ His co-speaker, Michel Jarraud, Secretary-General of the World Meteorological Organization, pointed to another highly condensed graph showing the comparison of temperature scenarios with two curves in blue and red climbing more or less steeply into possible climate futures (Figure 1b). He commented on the graph saying: ‘The children born now will *see* the last part of the graphic’ (my italics).²

During the press conference, when the IPCC scientists pointed to the highly abstracted and

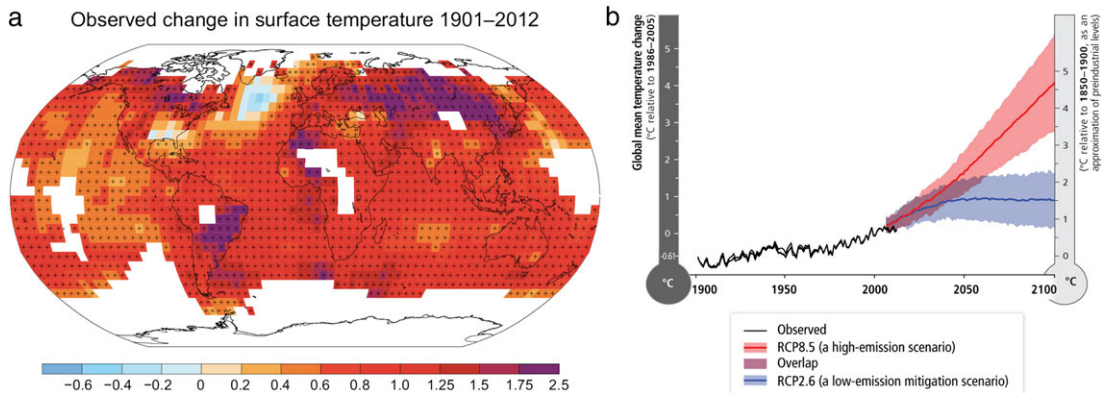


Figure 1 These figures were shown at the press conference of the IPCC when presenting their report in Stockholm on 27 September 2013. The graphs are part of the summary for policymakers of working group I (The physical basis). (a) Map of the observed surface temperature change from 1901 to 2012 derived from temperature trends determined by linear regression from one dataset (orange line in panel a). Trends have been calculated where data availability permits a robust estimate (i.e. only for grid boxes with greater than 70% complete records and more than 20% data availability in the first and last 10% of the time period). Other areas are white. Grid boxes where the trend is significant at the 10% level are indicated by a + sign. (b) Past, present and future in one chart. Shown are possible increases in the global average temperatures by 2100. The black line is based on measurements from the past: the red and blue line with the growing corridor shows the spectrum of possible futures. These are based on the RCP scenarios 2.6 and 8.4 relative to 1986–2005 and 1850–1900

Source: IPCC, AR5, Working Group 1 Figure SPM.1b (2013) and Working Group 2 Assessment Box SPM1 Figure 1 (2014)

conventionalised graphs, they communicated the essence of climate scientific findings using visual media. By speaking colloquially about the images, the scientists metaphorically related the data abstractions back to their concrete meaning for humans on Earth. This way of appealing to the public by way of visual aids has been practiced also at the COP21 in Paris at the end of 2015, where time and again scientific images published by the IPCC were employed to inform, convince and warn political decisionmakers. Once more the images chosen were whole Earth images, signifying in most cases the fate of humanity on this planet.

Invisibility of climate and media as environment

Scientific images fulfil several roles that differ greatly from other uses of images in visual culture. On the one hand they are tools of knowledge acquisition. On the other hand, they are the 'showplace of science' ('Schauplatz der Wissenschaft'; Heintz and Huber 2001, 34) where scientific knowledge aggregates and is staged into a visual form to be discussed and communicated.

The methodological framework applied here is German 'Bildwissenschaft' ('pictorial science'), the outward expansion of art history that bears differences to Anglo-American visual studies (Bredenkamp *et al.* 2015). The approach 'rests on the assumption that the choice of a particular pictorial form,

medium, or type has implications that hide in plain sight and inform the object of study and the manner in which it is studied' (Bredenkamp *et al.* 2015, 1). The analysis of pictorial media not only examines what was intended to be visualised but also tries to access layers of 'disguised symbolism' (Panofsky 1953) beyond the factual form, which materialises in any pictorial medium. It takes as a starting point methods of image analysis (iconography, iconology, methods of comparison and formal analysis), but also of media studies that analyse media as cultural techniques to process data (Kittler 1993). The perspective brings into focus the (technological) media conditions of possibility of knowledge but also of the media aesthetic of visual appearance which grounds media epistemology. In the case discussed here, specific visualisation methods of pictorial media are analysed in their relation to digital image technologies as the necessary framework for possible Earth knowledge and potential world views. Ecomedia like climate visualisations consolidate the dominant way of perceiving the environment: they add another layer to the basic concept of 'media as environment' (McLuhan 1964) as ecomedia as environment or image as environment. Environmental perception in the case of climate change research is based on media, therefore the perception of the environment is firmly rooted in media aesthetic.

A key function of scientific images is to 'make visible' (Rheinberger 2001, 57) complex processes and data

structures. In these terms, visualisation is nothing less than a method for making the invisible visible (Hacking 1983; Latour 1986; Rheinberger *et al.* 1997; Snyder 1998); it is this media-epistemic key function of visualisation techniques on which large parts of climate imagery are built on. Climate images are visualisations in exactly this literal sense: they make entities visible which otherwise could not become evident. Climate as an epistemic object is not something simply given but has to be constructed and mediated. The future is invisible; global temperatures are statistical entities that can neither be experienced nor seen. Many places and moments get statistically calculated to produce information about an average climate; most prominent is the 2° goal (the 2° ceiling on global warming thought necessary to avoid the most serious effects of global warming above pre-industrial temperatures). But otherwise atmospheric events, such as heat, air pressure or wind, are invisible also.

Because of the invisibility, climate, as an object of research, needs to be scientifically constructed and visualised with the help of instruments and graphs; there is no chance for scientists to learn about these phenomena in any way other than by media – the knowledge of science in large parts is homologous to the knowledge of media, based on measurements, networks and visualisation technology (Edwards 2010). It was only through new strategies of visualising numbers of observations and measurements that climate could become evident and, as such, an epistemic object of research.³

It is mainly scientific images that the different actors and recipients are currently inspecting when discussing possible climate futures. Therefore, it is important to draw attention to the very different layers of meaning these images obtain. In the following I will discuss how, by simply employing the colour range from blue to red and purple to mark temperatures that political implications and emotions (like fear of doomsday) are inevitably attached to scientific graphs and maps – and why it is impossible to get rid of dissonant rational and emotional reactions when confronted with scientific climate change imagery. As a consequence of my pictorial analysis, I will discuss the imagery of burning worlds in a cosmogrammatic framework derived from the approach by John Tresch (2014). For this purpose I will relate the climate world maps to the long tradition of cosmic views of Earth from outer space before and after the Apollo program, most prominently the Earth photographs by NASA 'Earth-rise' (1968) and 'Blue Marble' (1972). The images are not only an epistemological scheme for generating world knowledge and for transforming geo-space into an operational space, they also formulate a new cosmology of planetary future that constrain ontological questions.

By doing a media theoretical and art-historian analysis of the red maps of climate change the article

proposes to add to the developing research that has been done on climate change imagery throughout the last years (Hulme 2009; Manzo 2010; Doyle 2011; Grittmann 2012; Schneider 2012a,b; Mahony 2015) and stimulate conversations across researchers in geography, cultural studies, and science studies.

Scientific maps of global warming

One particular product of the scientific process of climate model data simulation became one of the most widespread icons of climate change: the image of a blue planet that has turned red. There are many examples that apply this way of depicting the earth and which prove that this type of climate world map has become conventional (Figure 2). Such pictures have been included in the reports of the IPCC parallel to the reports being printed throughout in colour since the year 2001.⁴

The climate report by the IPCC published in 2013/14 makes extensive use of world maps to present climate as a territorialised object. All maps are coloured in false-colour display, meaning that they show data with the help of green, red or blue. For my case study I have chosen visualisations of future temperature variations based on climate change scenario simulations generated by the German Climate Computing Centre (DKRZ) in Hamburg as their German contribution to the Fifth IPCC Assessment Report (Figure 3). The visualisations were carried out using a coupled atmosphere–ocean model developed by the Max Planck Institute for Meteorology (the MPI-ESM). The cartographers used Avizo Green software to generate the global visualisations. It may be reminiscent of Google Earth, which also places a so-called 'world-module' with the established default settings of global views at the centre of its interface. On the website of the German Climate Computing Centre, visitors can compare global views of different climate scenarios – stagnant, or as a development into the future; they also can watch the Earth heating up in a 3-minute clip.

Although within the IPCC reports these maps are given in an egg-shaped so-called Mollweide-projection (an equal-area, pseudocylindrical map) as used in Figure 1a or 2, I decided to use the globe-shaped example because its form gives a clearer possibility to compare this 'visiotype' (Pörksen 1997) to the Blue Marble. This may allow, in particular, an exploration of the following question: What does it mean culturally if 'whole Earth images', 'images of wholeness' or 'global images' are used for exploring and communicating scenarios of changing climate futures? It is my thesis that the various red climate future world maps are perceived in connection and comparison to the collective cultural memory of the Blue Marble in the process of image perception. The Blue Marble, being the key image of the ecological awareness of Earth since its publication,

Annual mean surface air temperature change (RCP4.5: 2081-2100)



Figure 2 Annual mean temperature surface change for the scenario RCP 4.5, 2081–2090 calculated by different simulation models

Source: IPCC, Working Group I, AR5, Fig12–9

rolls out the horizon against which the climatic planetary visions are seen – even if they are reproduced in flat, printed reports using different types of map projections. Both image types, though they follow their own photographic and cartographic rules respectively, depict limited parts of Earth and share a global ecological claim. It is the fate of ‘our home planet’ (NASA) being depicted in the maps and the Blue Marble photograph alike.

I want to start by looking at what the red world maps tell us scientifically. At first glance Figure 3 appears rather simple and straightforward. Taking that into account, its simplicity is an extreme contrast to the complexity which lies behind. The visualisation shows six spherical views of a circular globe with Africa, Europe and the North Pole brought into the spectator’s range of vision. The figures demarcate pair by pair changes

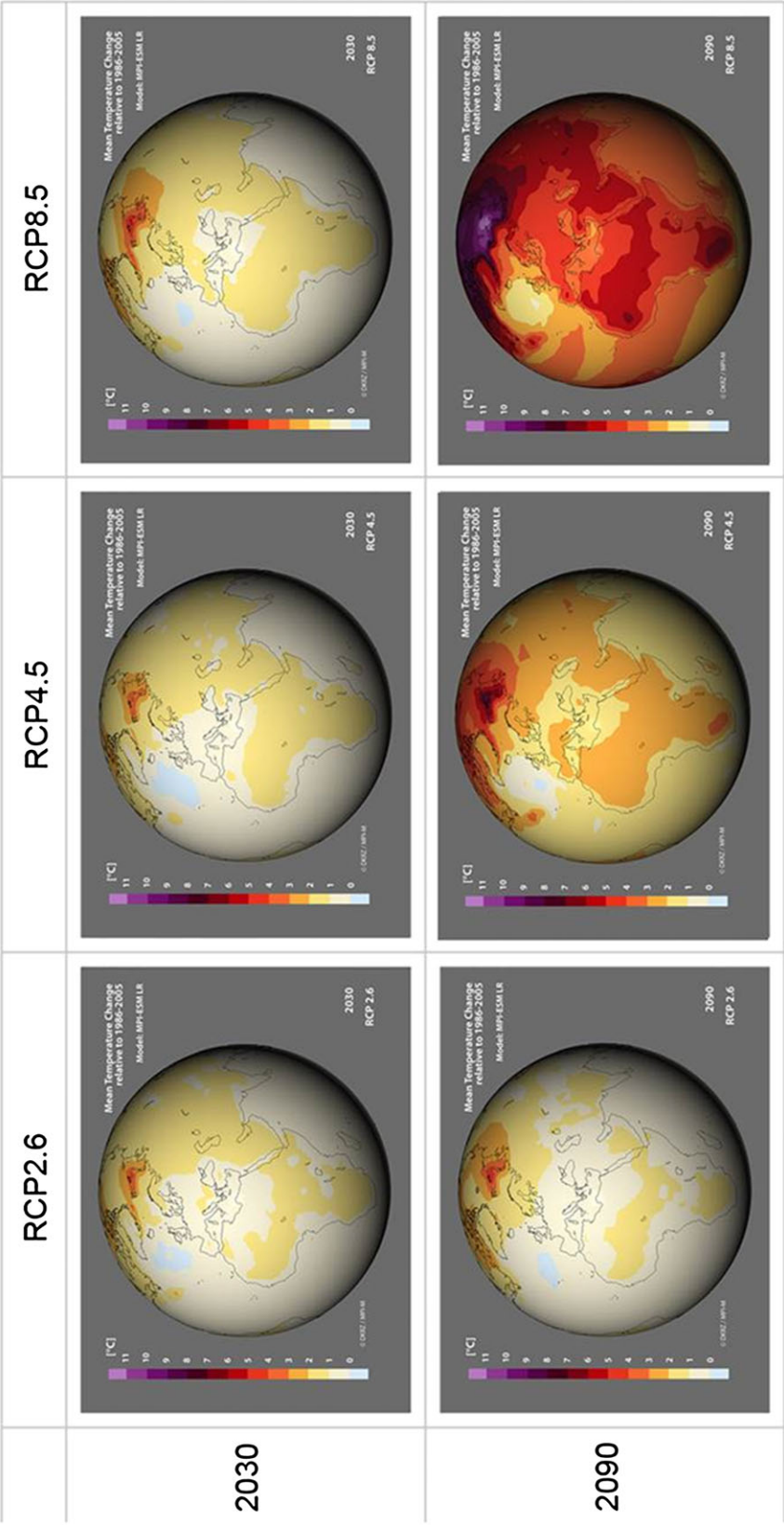


Figure 3 The figures show simulated changes in air temperature for three scenarios based on Earth system model runs for the years 2030 and 2090 in comparison. The model runs are part of the German contribution to the Fifth IPCC Assessment Report of 2013. For a full resolution zoomable image of each global simulation see the website of the DKRZ

Source: <https://www.dkrz.de/Klimaforschung-en/konsortial-en/ipcc-ar5/ergebnisse/2m-temperatur-en>, accessed January 2016

in air temperature close to the globe's surface for three different scenarios based on the ensemble averages of the climate model simulation runs for 2030 and 2090. By comparing the globes, it becomes possible to observe the process of temperature change in a manner similar to that of a sequence of single shots; viewers are expected to fill in the gaps between the pictures. The colour map, using false colour-coding ('false' meaning that the colours are not the 'real' colours of the object) given in the legend, is adapted to the common intuitive colour symbolism of cool (blue) and hot (red). In this case, a temperature range from -1°C to $+12^{\circ}\text{C}$ is used, depicted by a colour gradient ranging from blue through yellow to a light red, a dark red and three shades of purple. Colour hue is arranged symmetrically while gradually changing in luminosity. The colour blue only appears in tiny spots on the maps because there are no decreasing temperatures in these scenarios. A light purple symbolises the highest temperature increase.

The spherical views of the planet present two time steps within each storyline, the so-called RCP (representative climate pathways) scenarios. The IPCC scenario storylines bear an explicitly normative view of possible futures. The scenarios displayed here outline a range of either cutting down greenhouse gases quickly and radically (RCP 2.6) based on an idealised assumption of rapid economic growth or on a balanced emphasis on all energy sources (RCP 8.5). To put it simply: RCP 2.6 can be called a best-case scenario, whereas RCP 8.5 is a worst-case scenario and more or less business as usual (this development very much extrapolates the greenhouse emission development of the last years).⁵

In contrast to climate future time series charts, like the ones from the current IPCC report (i.e. Figure 1b), isosurfaces⁶ – the red shaded zones patterning the globes – are used for the red globes to demarcate the division of temperature changes around the world. Figure 1b and 3 depict RCP scenario data in a very different way: while the curves make no statement on the geographical distribution of the temperature changes, the red zones are clearly territorialised. A statistical mean value of global temperature change is shown in its distribution across the planet as it varies for each region. By showing the geographical distribution of temperature, it becomes clear that an average global temperature change of 2° does in fact mean that even higher temperature increases can be expected on land, especially in the North Pole region, while ocean surface temperatures change less.

What is made obvious on a scientific level of reading the visualisation of temperature changes with the bright red and loud set of so-called false colours is the argument that this whole process seems to be highly abnormal in comparison to the average climate of the last generations. This becomes even more explicit if

one compares the 2014 version of global future maps to the German contribution to the Fourth Assessment Report. In the case of the previous reports, similar visualisations were produced, but the 2007 maps for futures scenarios, for example, used a smaller temperature range, ranging from -0.5° to $+8^{\circ}$ to indicate the worst-case scenario. In the latter, this spectrum ranged between 6°C and 8°C marked by different shades of magenta.⁷ For the latest calculations the temperature spectrum got extended and the colour spectrum was altered – now it is the highest temperatures from 8° to 12° ranging from red to purple and magenta. From one report to the other the future projections did not only become 'redder' but also more purple. At the same time, it appears that there is no consistent colour spectrum throughout the different generations of IPCC reports to mark certain degrees of temperatures, this holding true for future scenarios and temperature anomaly maps alike. With each report the colour scheme got adapted to the new findings.

The scopic regime of the planetary perspective

When climate science today makes use of a variety of global maps to explore and communicate findings, one has to question the broader and ambiguous iconography which informs this image, since it is the idealised or even mythical vision of the blue, vulnerable and perfect marble that is impaired by the climate future portrait of Earth with their application of the colours yellow and red (Figure 4). Interestingly, this type of fiery red climate planet got a nickname, which might prove the relevance of the image and provide a frame of reference beyond science. In informal speech and popular culture the climate heat maps got connected to an image category illustrating the metaphor 'burning worlds'.⁸

When the Blue Marble photograph was published in 1972, it immediately evoked reactions like deep respect, admiration, devotion, and a sense of awe (Jasanoff 2001; Diederichsen and Franke 2013; Greismühl 2014; Dunaway 2015). For these reasons, the image of the Blue Marble became one of the most widespread icons of the environmental movement – at present there seems to be no climate change protest march that does not make use of a giant blow-up of the Blue Marble. But the Earth's photograph from outer space was ambivalent from the very beginning (Cosgrove 1994). It not only depicted the vulnerability of life on the planet but re-staged the contemplation of the sublime and the intangible greatness of nature beyond all possible imagination. This contemplation, though, is full of longing for home and distance at the same time. The Blue Marble photograph therefore provided a new indissoluble tension to the relationship of humans and nature.

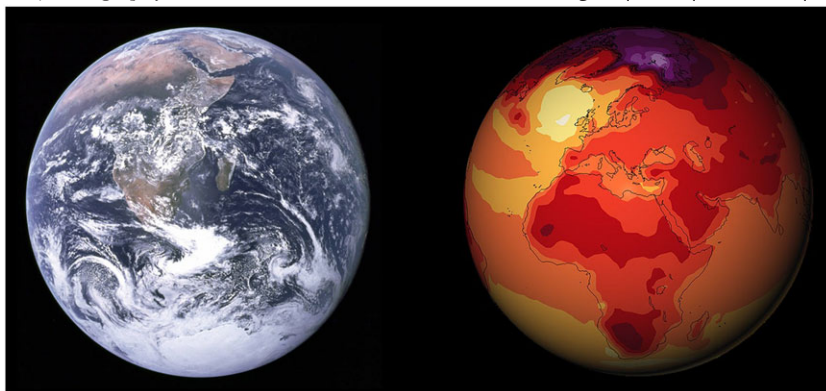


Figure 4 Collage from the Blue Marble, photograph of the Earth, taken by the crew of the Apollo 17 spacecraft in 1972 and a global map realised by the German Climate Calculation Center

Source: Blue Marble: image courtesy of NASA Johnson Space Center; Red Globe: DKRZ, 2014. Reprinted with permission from Ref 15. Copyright 2014 DKRZ

Building on the indissoluble ambiguity of the Blue Marble photograph, I would like to argue that both images, the Blue Marble and the burning worlds, share what Christian Metz termed a 'scopic regime' (Metz 1975), a term that Leon Gurevitch has developed further to analyse different types of 'digital globes' like Google Earth (Gurevitch 2013), his reading being applicable to the global grid of the Avizo Green climate visualisation software, too. Although the images are produced by very different methods and technologies, they follow the scopic regime of 'whole Earth images'. Both image types – analogue photographs of the Apollo program and the digital globes – create a 'theatrical platform' (Gurevitch 2013, 337) that stages Earth in a planetary perspective of wholeness by showing it floating in a black and void cosmic space.

By now there is a comprehensive body of literature about the media-aesthetic and political implications of the global scopic regime which has been analysed and criticised by many scholars as a modernist vision of the Earth as a globe involving geopolitical implications and technological superiority (Haraway 1991; Arendt 1992; Cosgrove 1994; Ingold 1995; Marksches *et al.* 2011; Parks 2013; Gurevitch 2013; 2014; Russill 2016). Most important is the criticism that the global view offers an artificially detached godlike perspective from beyond the planet, tracing back to a fictional Renaissance perspective. In that light, Earth does not surround the subject like an environment, but separates the human perspective from Earth (Ingold 1995). When digital globes today are 'the primary Earth representations of our culture' (Gurevitch 2013, 335), it is important to understand what is at stake, when the scopic regime of whole Earth images is even perceived as a naturalised mode of looking at Earth, although its modes of fabrication are part of an artificial and

mediated environmental perception (Russill 2016, 237). Here it is important to understand that whole Earth images are powerful actors of what Steward Brand and Al Gore called a 'war of perception ... over earth imaging' meaning that the perception of global environments are 'the sites of conflicts determining the future of human civilisation' (Russill 2016, 236, 242). On the other hand, the global images manifest how this sort of Earth perception is completely dependent on media aesthetics. The implicit consequence of this scopic regime is that the bird's-eye view of digital globes functions as a 'theatrical platform' for 'media events' (Gurevitch 2013, 327). The digital globe can again and again be 'reconfigured according to the environmental data that is fed into it' (Gurevitch 2013, 336), like updated climate data of global warming. At the same time, the scopic regime of whole Earth imagery indicates a process of alienation; it isolates observers from Earth. The global view is a product of the 'dialectic of enlightenment' when Earth got objectified with all the negative effects – objectification being a license for exploiting and controlling Earth's resources; it irreversibly separated nature from culture (Adorno and Horkheimer 2002; Gurevitch 2013; 2014).

Iconologies of burning worlds

To enter deeper into the iconology of the burning worlds, it is necessary to pose the questions Roland Barthes asked in his famous article on the 'rhetoric of the image': 'How does meaning get into the image? Where does it end? And if it ends, what is there beyond?' (Barthes 1977, 40). These questions are crucial because the cultural meaning of the scientific graph does not end with the decoding of the colour spectrum, limited to the concept of temperature means. What I want to argue here is that a new symbolic layer cannot

be circumvented when visualisations presenting climate science knowledge make use of representations that resemble the visiotype of the Blue Marble.

We have already learned that the most affected regions appear in dark red, purple and magenta, which on a level of everyday experience might remind one of a very bad sunburn. If we read the image as an x-ray picture for medical diagnosis, we can conclude that Earth does not look healthy at all. The beholders of this image are confronted with a damaged or even destroyed planet. It has lost its unique protecting and shielding atmosphere and instead produced a *torrid zone* now covering the whole globe; Earth does not look like the living planet – Gaia – anymore, which the Blue Marble photograph portrayed so impressively. Instead we can observe Earth transforming into a planet hostile to life or becoming a ‘dead planet’. Because of its red surface it might even remind one of Mars, the red planet that was named after the Greek war god.

Here, intuitively and implicitly, powerful narrations of the ‘end of the world’ can connect. The burning world’s image bear, intentionally or not, elements of horror and shock that, if we like it or not, are attached to the image. This is why the image triggers conflicting reactions that go beyond ‘pure’ scientific readings. Observers do not react rationally in the scientific ideal of being objective and without emotion. The colour scheme intuitively triggers an aesthetic of fear, urgency and alarm. What the blushing of the planet tells is the

story of an irreversible development towards a heating planet on which the climate machine has gone haywire. This though actually alludes to the traditional grand narratives of disasters and final endings. In the Christian world this would be the Last Judgment, Purgatory or the Apocalypse; but such end of the world views are also shared by other world religions. This has inspired artists to create the most fascinating and inventive pictures. From an aesthetic and artistic point of view, narratives of a final ending have led to some of the most impressive image creations in culture. The horizon of Western cultural imaginations of ‘grand narratives’ might appear like the impressive paintings of Hieronymus Bosch or Hans Memling (Figure 5). But the imagery of recent years is especially rich also in shaping imaginations about end of the world scenerios and ‘future as catastrophe’ (Distelmeyer 2013; Horn 2014), to name only a few: the movies *2012*, *Melancholia* or computer games like *The last of us*.

Colour symbolisms of red in art and science

In spite of all these disastrous and horrifying storylines burning worlds inspire, red and blue simply belong on a practical level to the intuitive colour scheme for temperatures. From a cartographer’s point of view, the blue-red colour scheme as such is without alternative. This is what a climate cartographer who produced the map for the German Climate Computing Centre



Figure 5 The Last Judgment, triptych by Hans Memling, 1467–71, National Museum Gdansk. On the right-hand panel, the damned are being dragged to the burning Hell

Source: Public Domain

assured me.⁹ A historical perspective on the colour-coding of different temperatures also follows this argument, as the scheme has a long tradition in the history of cartography and data visualisation. Climate zones have been drawn in this way (Figure 6) for more than 200 years. Blue is a cold colour, red is a warm colour, so it is easy to see how the scheme derived from nature and experience. When only rising temperatures are calculated in the models for possible climate futures, shades of red colour are the only way, from the creator's point of view, to depict this outcome – green would contradict this essential information. So what do standards of map design tell us about the application of colours?

Since the early twentieth century, guidelines exist at a level of practical knowledge about the use of colours in a 'good' and 'effective' way and how to apply this to data graphics and maps (Bertin 1983 [1967]; Tufte 1990; MacEachren 1995; Palsky 1996; Stauffer *et al.* 2015). As data visualisers know, graphics are capable of containing only a limited amount of information. Here, over time colour became a very effective way of

enlarging the spectrum of information that could be put into a single figure because it allowed much greater selectivity of perception compared with black and white.

Graphic designer and author Edward Tufte very clearly warned against the use of colour for data without reflection, stating that '[t]ying color to information is as elementary and straightforward as colour techniques in art', but 'putting a good colour in a good place is a complex matter. Indeed, so difficult and subtle that avoiding catastrophe becomes the first principle in bringing colour to information: *Above all, do no harm*' (Tufte 1990, 81). With this clear statement, Tufte attributed a certain risk to the adding of colour to figures – the risk that using colour indiscriminately might even spoil the communicative aim of a picture. Tufte gives a citation of the Swiss cartographer Eduard Imhof (Imhof 1982), which can very much be applied to the 'burning worlds' example.

First rule: Pure, bright or very strong colours have loud, unbearable effects when they stand unrelieved over large

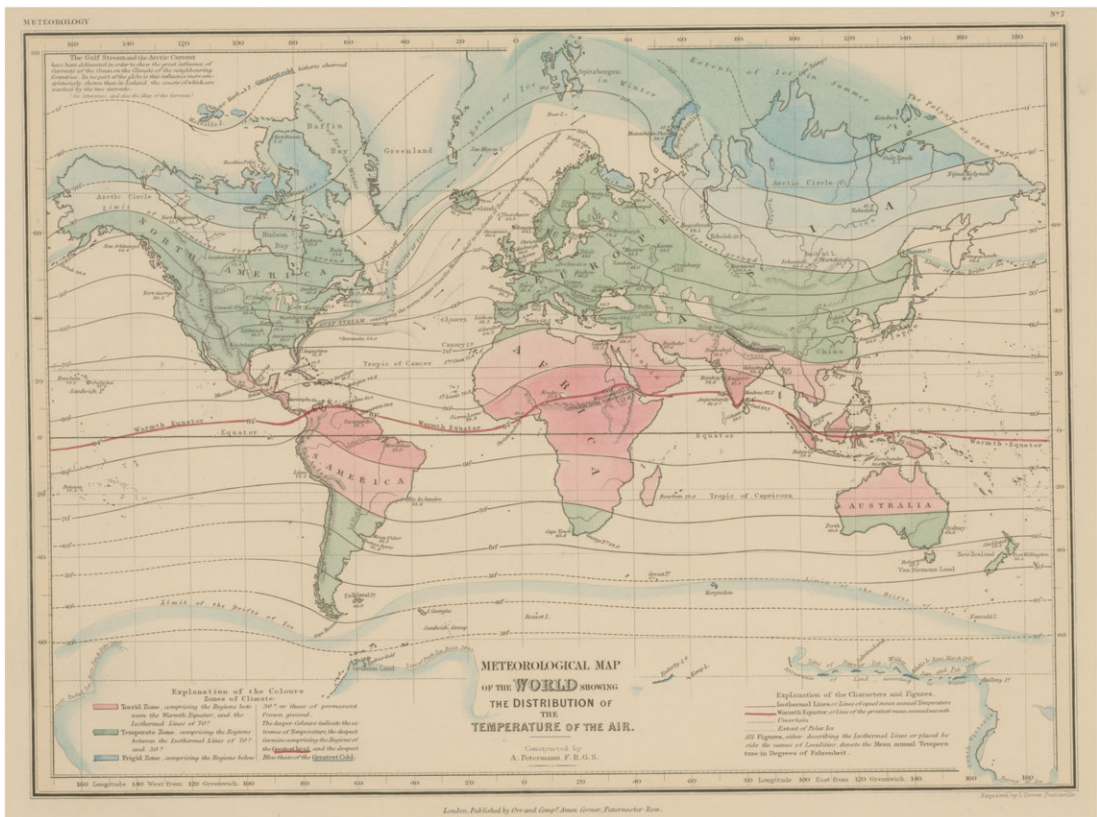


Figure 6 Early climatic world map from the Atlas of Physical Geography published by August Heinrich Peterman in 1850. The map marks climate zones by color to illustrate the distribution of the temperature of the air. The torrid zone is colored in red, the temperate zone is colored in green, the frigid zone is given in blue

Source: Staatsbibliothek Berlin – preußischer Kulturbesitz

areas adjacent to each other, but extraordinary effects can be achieved when they are used sparingly on or between dull background tones ... Noise is not music. Only a piano allows a crescendo and then a forte, and only on a quiet background can a colorful theme be constructed. The organisation of the earth's surface facilitates graphic solutions of this type in maps. Extremes of any type – such as highest land zones and deepest sea troughs, temperature maxima and minima – *generally enclose small areas only*. If one limits strong, heavy, rich, and solid colors to the small areas of extremes, then expressive and beautiful patterns occur. *If one gives all, especially large areas, glaring, rich colors, the pictures have brilliant, disordered, confusing and unpleasant effects.*

Imhof (1982, quoted in Tufte 1990, 82, my italics)

In the case of climate future maps, this is exactly what happens. In this sense, the 'burning world' maps are to be described as noisy, brilliant, disordered and confusing with unpleasant effects. By applying maximum colour intensities with varying saturations and not, as Stauffer *et al.* suggest, the more moderate HCL colour space with balanced colour saturations, such maps can be rated 'misleading and distorting' (Stauffer *et al.* 2015, 205). On the other hand, one might make the point that in the red world maps the colouring clearly parallels the finding that global warming does not affect small areas only but rather the whole planet. Therefore, the cartographic outcome has to be highly unpleasant, even if it spoils the standard of 'good cartography'.

Another general recommendation on how to apply colour in information design is that it should rely on logic which bears similarity to nature (Tufte 1990, 90; Palsky 1996, 134). This advice cannot be applied to the use of red, because it is a signal colour that occurs in nature only rarely. Still, the coding of red as heat and blue as coolness is 'natural' and as such very successful, precisely because pictures encoding red and blue for temperatures are perceived at an intuitive level.

What is of interest here is that beyond the natural coding of red and blue to depict temperature, very different and maybe even stronger symbolic meanings are inevitably assigned to the subject which likewise belong to the established semantics of red and blue. A cultural history of colour here offers many other and again conflicting paths of interpretation (Gage 1993; 2000). Red became conventionalised as the colour of highlighting, emphasising, appealing and of marking danger and action at the same time, because red also stands for the destructive power of torrid heat. Therefore, in information design, red triggers very different symbolic layers such as danger, warning signal, intensity and the deviation of a norm – especially in the case of visualising global risks.

These very different meanings of the red colour have to be weighed up because of the potential problem of misdirecting attention and creating confusion. So, by using the conventionalised colours to code temperature in the case of climate future scenarios, very different and uncontrollable meanings are liable to be incurred. Even though the application of colour scales to maps is by itself a scientifically objective signal that states its own truth for visualisation environments, what these signals actually mobilise in different recipients usually is a much broader response, depending on cultural, national, practical or aesthetical knowledge, traditions and experiences.

The polyphony of meaning is something that has been studied extensively in the field of semiotics, philosophy and art history (Hatt and Klonk 2006). Artists and creatives in advertising, who target a broader audience, are very aware of the fact that the meaning of images cannot be restricted completely. There always will be a *surplus* of uncontrollable meaning. In the case of expert graphs, this means the pure informative level of the intensive colour red in order to code warming is very strongly directing attention to all these other notions of red in connection to the motive.

Influencing decisions or just decision-making tools?

One could argue that even campaigners from a non-governmental organisation like Greenpeace could not have done a better job than the cartographers and computer graphic designers in the international research labs of climate research. The visiotype of burning worlds works as it fulfils all requirements of a strong, intuitive and iconic image. It similarly addresses the rational *and* the emotional. Moreover, the message is straightforward: global warming is an alarming problem people/we/politicians have to face. The image is about changing the present in order to never witness this future. Politicians need to act to change what the pictures show, because burning worlds are beyond imagination.

Here the political role climate science obtained when discovering global warming has to be clarified. Today, climate images are political images (Walsh 2009; 2013; Schneider and Nocke 2014; Mahony 2015). The argument of Gilles Palsky that colour transforms maps into propaganda when used to depict political regions (Palsky 1996, 133)¹⁰ becomes strangely valid when applied to climate change. Because it has become impossible for research fields like climate sciences, which address global risks, to work only in the scientific realm. Instead, in the field of climate change research the boundaries between science, society and policy are eminently blurred because the issues

climate sciences address are in the centre of great socio-political and economic concern. IPCC reports are written by scientists to inform policymakers before they take decisions. Therefore, climate scientists in this field find themselves in an area of tension between different role models, because they are expected to communicate the political meaning of their findings to policymakers but also to the public (Pielke 2007)¹¹. It certainly has been an unintended consequence of role expectations towards scientists and the scientific, but also fiery red, colour spectrum that some 'climate sceptics' saw the visualisations as an indicator for IPCC scientist being 'alarmist' (Walsh 2013, 181); that is, following a hidden political agenda. Such reproaches seem to neglect what Barthes and many other scholars argue: there is nothing like a 'pure representation' (Bredekamp *et al.* 2015). The red global maps not only communicate pure knowledge, pure numbers, models or findings – a kind of rational objectivity science would tell us without distortion. Here, the myth of photographic 'naturalness' returns in form of the myth of *pure data visualisation* considered as an act of 'natural inscription' (Walsh 2014; 2015) meaning the ideal of an undistorted map, a map, that only and directly shows what is in the numbers and nothing beyond. But maps always are subjective; they include many subjective decisions such as the choice of colours, segments and different types of map projection. In other words, cartographers have to make aesthetic choices that influence the message.

Images narrate in a particularly striking way how the spheres between politics and science become blurred (Walsh 2013; Mahony 2015). The colour red in climate change expert graphs is not only used to highlight rising temperatures, but at the same time brings forward the message of an imminent climate disaster. Even though climate scientists might follow the IPCC ideal to be only 'policy relevant' and not 'policy prescriptive', one can trace the inseparable intermingling of facts and values (the 'is/ought-divide'; Walsh 2009) amongst the scientific images simply by looking, for example, at the colour red, which at the same time is the colour of disastrous news. Therefore, the warning and hot colour red here highlights not only pure temperature increases but can be interpreted, similar to another prominent IPCC diagram called 'Reasons for concern' (Mahony 2015),¹² as highlighting the inescapable political relevancy of climate research findings.

Affects of shock images

Here we can return to Roland Barthes again and use his approach to the rhetoric of the image to analyse in more depth the iconography of global warming (Barthes 1972).¹³ Barthes coined the term 'shock photo' to identify the rhetoric whilst analysing photographs. The concept was taken up by Susan Sontag

and developed further (Sontag 2003). Although Barthes used this term to speak about press photography in a more general sense, the term can also be applied to the 'cartographies of danger' (Monmonnier 1997) such as the case discussed here. The shock image is intended to catch the attention of the recipient without allowing them to reflect on it. This leads to Barthes' thesis that '[m]ost of the photographs exhibited to shock us have no effect at all' (Barthes 1972, 116). Like the aesthetic concept of the sublime, horror relies on the fact that the beholder is looking at the image from a secure (distant) position. What is signified in the global climate maps is too vast in space and time to be imagined and to produce long-term reactions. On one hand, the sunburned data images of Gaia are meant as a call for action, as they display the greatest possible moral failure of humankind. On the other hand, they portray climate change as a cataclysm in which human beings are condemned to fatalistic powerlessness, even though the reason for the heated planet is anthropogenic, and as such, induced by culture. Therefore, if one searches for a politically engaging and activating image, burning world imagery might not be successful in the long term because fear does not lead to action, as Nicholson-Cole and O'Neill found out in an empiric study on climate imagery perception (Nicholson-Cole and O'Neill 2009).

The burning worlds are far beyond the conceptual grasp. Therefore, it is inconceivable for anyone to understand what the accelerating disturbances of natural order by human impacts will mean for culture. What is triggered though are, following Lorraine Daston, emotions such as horror and terror (Daston 1998). The reports of the IPCC are filled with red planets. By making such intense use of this global data picture type, the globes not only mark the next step of total Earth knowledge, they mark the failure of the controllability myth of the domination of humanity over nature. Human destruction of the shielding biosphere itself is made visible with the red world maps. The world-object, which becomes reshaped even more dramatically by computer simulation and graphics in every report, simultaneously delivers an allegory of the 'tragic triumph' (Schellnhuber 2010) of modern science—referring to the notion of the Greek tragedy in an uncanny way: knowledge about the catastrophe is achieved only at the moment of falling – evoking emotions of pity and fear but without any promise of relief (catharsis). The visualisations offer humans – the main actors of the tragedy – a sad insight as to where global exploitation has brought them and what future sufferings they will face. The observers of these modern devotional pictures are left with a feeling of vast impotence and can become prey to all sorts of apocalyptic narrations caused by their own actions like Judgement Day or Purgatory (Figure 5), which are deeply routed in the cultural memory.

Cosmograms of the Anthropocene

Images of global climate futures are generated by digital software like Avizo Green on the grid of a digital globe (the 'world module'), this grid being the stage for Earth's history and future. The visualisation software Avizo Green was originally developed to diagnose human diseases. If today it is used to diagnose anthropogenic disorders of the planet Earth and their possible effects on humans, here a drama of human actors comes into play.

If the Blue Marble image visually materialised a shift in public consciousness on the nature of the environment, one can ask what is the subsequent argument that is being summarised in the burning worlds today? I want to bring up a cosmic and cosmological perspective here. The 'planetary turn' that threw a gaze from the spaceship Apollo backwards at Earth has set up the framework to think and imagine ecological interconnectedness and boundaries since the 1960s and 1970s. Today the digital globe is the 'theatre of environmental spectacle' (Gurevitch 2013, 336) that stages different dramatic futures in the same visual format, although the cosmic gaze in the tradition of the space program turned spatially backwards while today this gaze is connected to a prospect into the future: cosmic futures in the rear-view mirror.

In conclusion and in placing red future world maps within the context of a universal narrative, I would like to use the term *cosmograms* in the burning world's discussion, adapting the general argument of historian of science and anthropologist John Tresch and applying it to red climate maps (Tresch 2014). Cosmology is defined as a certain worldview that tells the fate of the world (often by gathering and interpreting astronomical data), the origin and evolution of the universe but also its eventual fate – the Earth's fate being the fate of all human cosmologies. A cosmogram is a schematic figure depicting a cosmology, often created for the purpose of meditation. Even though the data maps of possible climate futures have not been constructed for meditation in the first place, switching to a cosmological perspective here is productive. Regarded as cosmograms, the red maps can be seen as representing a modern scientific 'materialized cosmology' (Tresch 2014, 164), designed by 'visioneers' (McCray 2013).¹⁴

Read as a cosmogram, the future map projections materialise and visualise the worldview of the Anthropocene, this concept of geology and deep time (Steffen *et al.* 2007) both made productive (Trischler 2013), critically reflected upon (Manemann 2014; Visconti 2014; Castree 2014) and challenged geophilosophically (Yusoff 2013) as the new universal narrative by many humanity scholars today. This final anthropocentric cosmology in many cases is visually argued within a geographical scheme that relates

human actions to their geographical environment (Castree 2014, 464), one of them being the red world maps. It regards humans as inevitably marking their imprint on the face of the Earth, climate change being considered as another geological human trace. Therefore, if we take these images seriously as cosmographic meditations, mirroring those that exist on the creation of Earth, it is even more important to ask questions that reach well beyond scientific iconography. Following John Tresch, cosmograms are practices of visual world creation; they reveal imaginings of order and chaos, future and origin, they tell where *we* go and where *we* come from. By doing so, they also shape imaginings of the future, of Earth and the Universe in its totality. Tresch claims that:

material cosmologies set the stage for cultural contacts. Western science was not a neutral 'tool of empire', using value-free facts to accomplish utilitarian ends; it carried culturally specific values ... and sought to redraw the map of the world, both metaphysically and geographically.

Tresch (2014, 166)

Read as modern cosmograms, climatology situates humanity in the following cosmological narration, which bears the mythical order of a triptych comparable to the winged altar of Hans Memling (Figure 5). Salvation (left panel): the Blue Marble is the world *we* came from; it recalls the past we can never return to, similar to the Garden of Eden, but also the hope for future salvation, if humankind radically changes habits to make a better (less carbon dioxide intense) living (Figure 4, left globe). Last Judgment (middle panel): scientists are currently talking about the ecological sins of industrialisation (Fall of Man). This is the age of detectable human impacts on Earth (the Anthropocene): the world is already transformed by humans to such an extent that the end of humanity is contained in the present (Figure 1a). The Damned: the right panel of the triptych would be the burning world, representing the devastating future of the worst-case scenario – the fate of the world being business as usual (Figure 4, right globe). Interestingly, this general structure of narration also underlies Pope Francis' 2015 ecological Encyclical (*On care for our common home*).

I am aware of the danger that my interpretations might sound cynical, pessimistic and even over interpreted, but this is not what I am aiming for. Since all future scenarios are fictional (not to be confused with unrealistic), as Elena Esposito argues, they all have the status of *possible* futures (Esposito 2007). Nobody knows how exactly global warming will develop and how this will change the world we know. Therefore, the burning worlds not only mark a step of scientific

Earth knowledge, they are blueprints for the Anthropocene worldview and the fears connected to it – nothing more and nothing less. The maps present a wish for planning capability, controllability, although at the same time they depict a world changing into chaos as the verb ‘burning’ already suggests. They display a wish to control what is out of control. They display the shock most people are unable to grasp, namely that humans and their technological systems are part of this image.

Being thus regarded, it also applies to the totalised images of climate futures that ‘cosmological ideas take on realist force when they are anchored, housed, and transmitted objects, technical networks, routine practices, and social institutions’ (Tresch 2014, 162). It is for this reason why it is so important to approach such images critically because of associated totalising concepts and holism, although it will be very difficult to enrich and complement the predominant scientific understanding of the world and find alternative ways to interpret and materialise the fate of the World.

Images are blueprints to imagine and shape reality. What a cosmogrammatic and cultural reading of scientific images might point out: we also need to find cosmograms as an alternative to scientific maps as scientific knowledge leaves a blank space that science alone cannot fill. There is a need for ways other than the cold inventory of heat maps that help to think, decide, imagine, narrate and envision the future – other ‘cosmograms’ – local or glocal, religious, political and cultural. A manifold of imaginings is urgently needed, imaginings, which are able to add cultural layers of meaning to the fate of the world we are facing in the red climate world maps.

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Notes

- 1 The press conference was held in Stockholm on 27 September 2013 to present the Summary for Policy Makers of the Working Group I contribution to the Fifth Assessment Report. The quotes were taken from the webcast.
- 2 *Ibid.*
- 3 This is what Alexander von Humboldt did in 1817 by using weather data to construct his famous map of climate zones (Schneider, 2012a,b; see also Regnaud's contribution in this issue). Because of the dominant construction of

- climate change as a scientific entity – and the invisibility paradigm which results from this definition, claiming that climate change as such is inaccessible directly to the senses and science is the most prominent authority to make visible climate change – the question of visibility has been discussed as a political issue (Knebusch 2008; Rudiak-Gould 2013).
- 4 The reports from the years 1990 and 1992 already contained limited pages with coloured maps. In 1990, eight pages of the 414 pages of the First Assessment Report of Working Group I were printed in colour. Here the colour spectrum from yellow to red, to illustrate global warming of surface temperatures, was used for the first time applied to a cylindrical equal-area projection map (Figure 5.4 of the report).
- 5 This production of predictive knowledge in the form of ‘scenarios’ and ‘storylines’ has shaped new scientific cultures that might be analysed under the term ‘cultures of prediction’ as Matthias Heyman suggests referring to Gary Fine's book *Authors of the storm. Meteorologists and the Cultures of Prediction* 2007. An approach towards cultures of prediction take into account the historically changing conditions and the making (practices) of these predictions.
- 6 An isosurface is a three-dimensional analogue of an isoline that visualises points of a constant value like temperature within a volume of space.
- 7 See Figure SPM 6 in IPCC AR4.
- 8 ‘Burning world’ is a common category in stock image data bases, like Getty, to illustrate global warming. The double meaning of ‘burning’ as in heat and burning fossils are found in book titles such as *How to stop the planet from burning* by George Monbiot (2007), and in science fiction series taking place in a post-climate-catastrophic world such as *Aviator: the burning world* by Gareth Renowden (2012).
- 9 Interview with Michael Böttger on 29 June 2011, cartographer and visualiser at the German Climate Computing Center, Hamburg.
- 10 Political colours: left wing, right wing; Communist blocks. Or in the USA, the recent colour scheme of red states (republican) and blue states (democratic).
- 11 Pielke categorised different ideal role options in the relations between scientists and their social world like the ‘pure scientist’ and the ‘science arbiter’ in contrast to these traditional roles the ‘issue advocate’ and the ‘honest broker of policy alternatives’.
- 12 There is another prominent IPCC diagram applying a similar colour spectrum from white, yellow to red called ‘Reasons for concern’. In this case, dark red explicitly highlights the most ‘dangerous’ climate change. It later received the nickname ‘burning embers’ and was left out of the 2007 report for its obvious meaning. Martin Mahony did an in-depth analysis of this graph, resulting in a parallel argument concerning the political and expressive meaning of colour. See Mahony (2015).
- 13 In his known analysis of the Panzani advert, Barthes differentiated three levels of messages which in different ways employ denotation and connotation: the first is the linguistic level of the message, the written text. It already contains a denoted and a connoted message. The second is the symbolic message; it is coded and iconic and tells

what the image might stand for. The third is the literal message, it is not coded. In the advertisement the denoted layer of information naturalises the symbolic layer. In contrast to photography, the red planets do not depict a reality that took place in the past, on the contrary it reverses the relation of time and space by depicting possible future events. 'The literal images leads to the scandal of horror but not to horror itself' (Barthes 1957).

- 14 Visualisers today call themselves visioneers, bringing together the terms 'vision' and 'engineer'.

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